EE4-16
EE9-AC6

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE PART IV: MEng and ACGI

ANALOGUE SIGNAL PROCESSING

Thursday, 13 May 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P. Georgiou
Second Marker(s) : E. Rodriguez-Villegas

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Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n *: Weak Inversion Slope factor.*

g_m*: Transconductance.*

1. a) Discuss the benefits of both analogue and digital signal processing with regards to power consumption, computational capability and resolution.

[6]

b) The differential pair transconductor shown in Figure 1.1 has MOS transistors which are operating in the weak-inversion region of operation with a slope factor n = 1.23.

Show that the output current $I_{out} = I_1 - I_2$ of the differential pair transconductor may be given by equation 1 and derive the transconductance efficiency. Show all your steps in derivation.

$$I_{out} = I_{tail} \tanh\left(\frac{V_{GS1} - V_{GS2}}{2nU_t}\right) \qquad (1)$$

[4]

c) Figure 1.2 shows a cascade of amplifiers where each stage uses the circuit from Figure 1.1. The total noise contribution in each stage is due to the two input devices of the differential pair and the gain of each stage is given by $G_i = -g_m R_{out}$, where g_m is the transconductance of the differential pair in Figure 1.1 and $R_{out} = 100\ G\Omega$.

(i) Given the bias current I_{tail}=20nA, calculate the minimum area A_1 required by the first stage, such that the total contribution of the noise at the output from the first stage is less than 500mV. You may assume $K_w = 1x10^{-30}\ V^2C$, $K_f = 3.5x10^{-21}\ Hz/F^2$, ΔF =100 Hz, f_h = 100Hz, f_l = 1Hz and the transconductance is constant over a given input voltage.

[6]

(ii) Propose suitable gate widths and lengths for transistors M1 and M2 for a typical 65nm process.

[2]

(iii) Given that the system shown in Figure 1.2 operates on a Bluetooth signal at 2.4 GHz, but conveys audio information at 44 kHz, propose one method to reduce the power consumption requirements before quantizing the analogue signal.

[2]

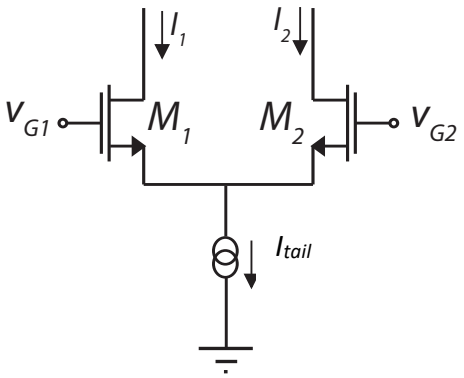


Figure 1.1

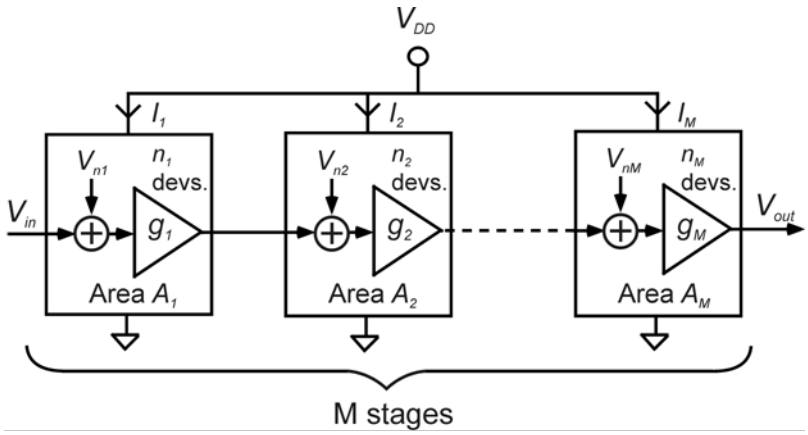

V_{DD}
I_1
I_2
I_M
V_{n1}
n_1 devs.
V_{n2}
n_2 devs.
V_{nM}
n_M devs.
V_{in}
g_1
g_2
g_M
V_{out}
Area A_1
Area A_2
Area A_M
M stages

Figure 1.2

2. This question concerns translinear circuits.

a) Explain why MOSFETs operating in weak-inversion make suitable translinear elements and state two limitations when using these in translinear circuits which may apply specifically to weak inversion operation.

[4]

b) Figure 2.1 shows a translinear circuit.

(i) Derive the transfer function I_{out} of this circuit using the translinear principle. You may assume that currents I_1 and I_2 are input currents and I_4 is a static bias.

[5]

(ii) Write down an expression for the output current I_{out} when I_{in}= Asin(ωt) and I_1 and I_3 are derived as:

$$I_3 = \left|\frac{dI_{in}}{dt}\right|, \qquad I_1 = \left|\int I_{in}\, dt\right|$$

[4]

c) Synthesise using Type A translinear cells, a multiplier circuit that implements the following function:

$$z^+ - z^- = (x^+ - x^-)y$$

Draw the complete circuit including all biasing schemes and remove redundant loops.

[5]

d) Explain one advantage of a Gilbert gain cell over a conventional translinear multiplier.

[2]

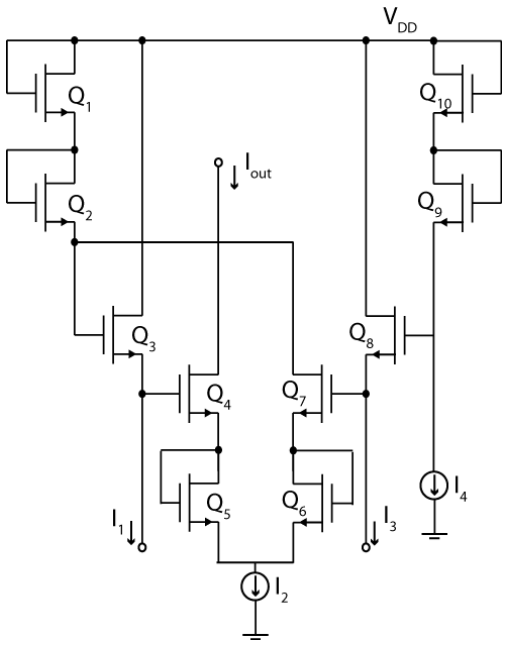


Figure 2.1

3. a) Explain the principle of companding, illustrating how the dynamic range changes through the signal processing chain.

[4]

b) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_0 \sinh(V_c/nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f(x) composed of just I_{in}, I_1 and I_{cosh}, whereby $I_{cosh} = I_0 \cosh(V_c/nU_t)$.

[6]

c) The transfer function of an oscillator may be defined in state space representation by the following equations:

$$\dot{X}_1 = -\omega_0 X_1 + \omega_0 X_2$$
$$\dot{X}_2 = -2\omega_0 X_1 + \omega_0 X_2 + \omega_0 U$$
$$Y = X_1$$

where Y is the output, U is the input and X_1 and X_2 are the states.

(i) By using the mappings below show how these linear equations can be mapped to non-linear log-domain design equations. You may use the following mapping for constant currents, $I_1 = I_2 = I_\omega$ and $I_U = I_O$.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U = I_U \exp\left(\frac{V_U}{nU_t}\right)$$

[5]

(ii) With these log-domain design equations, sketch a schematic of the final log domain filter using weak inversion MOS transistors.

[5]

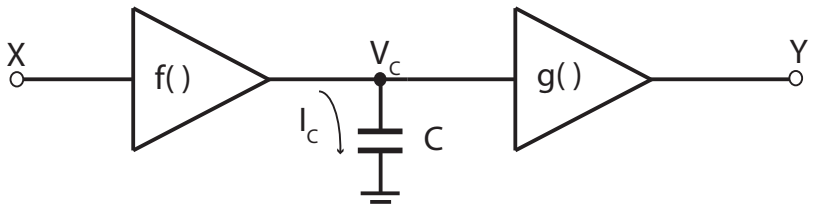


Figure 3.1

4.

a) Figure 4.1 shows a conventional second-generation positive current conveyor, CCII+.

(i) Explain it's operation principle and describe how impedances at each of its three ports, X, Y, Z, differ from a standard operational amplifier.

[3]

(ii) Draw the circuit of a transistor implementation of a bi-directional CCII+ which does not use any opamps.

[3]

(iii) Using current conveyors design circuits which implement:

- A grounded inductance.
- A full wave voltage mode rectifier.

[4]

b) Derive the generalized equation for the output currents, I_{outi} of the circuit in Figure 4.2 and explain the function of this circuit.

[4]

c) Figure 4.3 shows a circuit which is used as a variable OTA. Using your results from part (b) or otherwise derive an equation for the output current I_j as a function of the differential input voltage (E_j-V). You may assume all transistors are operating in weak inversion.

[6]

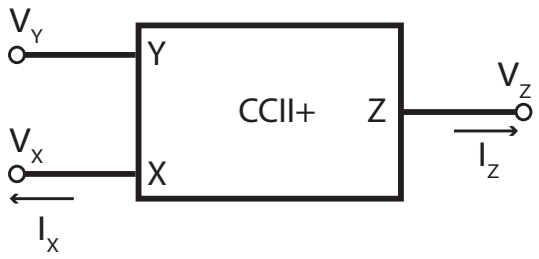


Figure 4.1

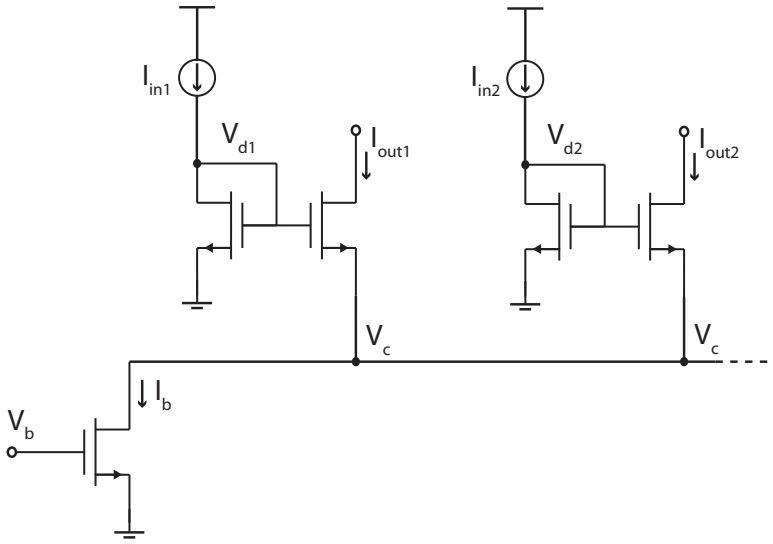


Figure 4.2

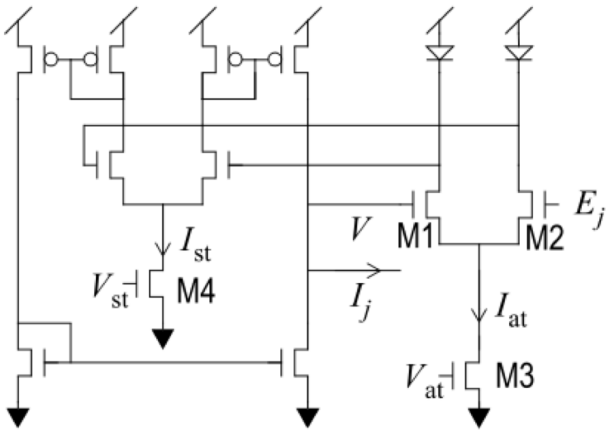


Figure 4.3

5.

a) Explain the principle of autozeroing with the aid of diagrams.

[4]

b) Explain how autozeroing affects the noise spectrum of the signal.

[4]

c) Figure 5.1 shows a switched capacitor amplifier correlated double sampling amplifier.

(i) Derive the output voltage of this amplifier V_{out} for both phases Φ_1 and Φ_2 showing how this removes any offset in the amplifier. You may ignore feedback capacitor C_{dg}.

[4]

(ii) Show how the circuit of Figure 5.1 can be modified to reduce the high slew rate requirements, and derive the change in output voltage after this modification.

[4]

d) Derive the output current, I_{out} for the switch current algorithmic cell given in Figure 5.2 showing how this eliminates any error caused due to the switches.

[4]

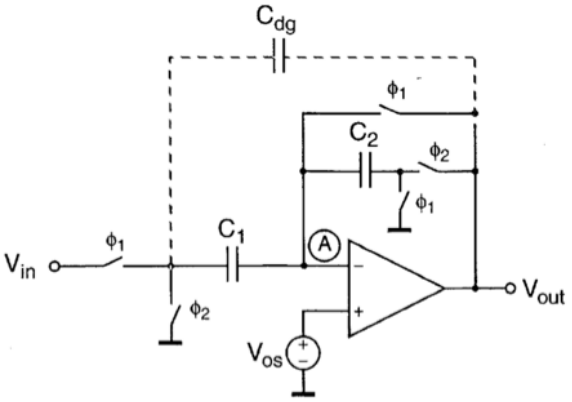


Figure 5.1

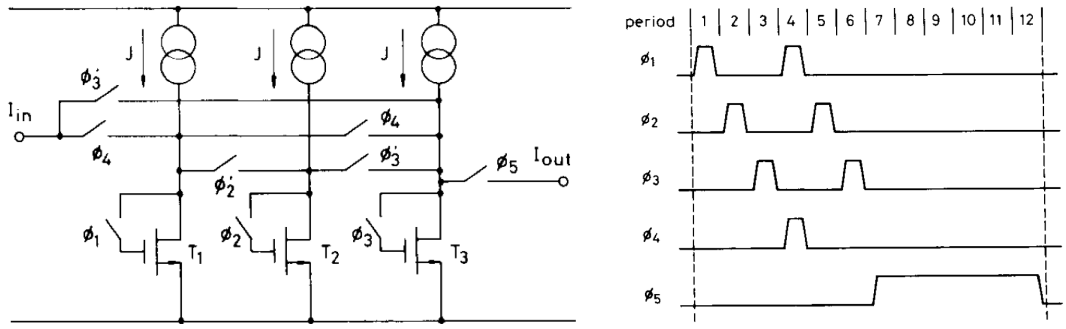


Figure 5.2

6. Figure 6.1 shows the block diagram of a cochlear prosthesis.

a) Explain the concept of dynamic range mapping used in this system and why it is beneficial.

[5]

b) Sketch the schematic of a suitable low-power Analogue Front End (AFE) which works for audio frequencies and explain its operation deriving its transfer function.

[5]

c) Sketch the schematic of a suitable logarithmic analogue to digital converter which also uses autozeroing to remove any offset in the amplifiers and explain its operation showing timing diagrams and equations where necessary.

[5]

d) Sketch a suitable schematic for the scanner block and explain it's operation.

[5]

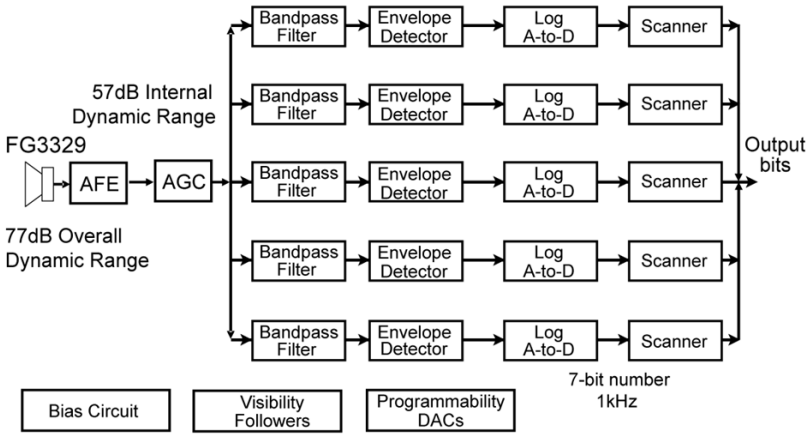


Figure 6.1